INCH-POUND

MIL-DTL-83725/15C
22 September 2016
SUPERSEDING
MIL-R-83725/15B(USAF)
30 September1983

DETAIL SPECIFICATION SHEET

# RELAYS, VACUUM, SPDT, 75 AMPERES RMS 15 KILOVOLTS (PEAK) OR DC

This specification is approved for use by the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and MIL-DTL-83725.

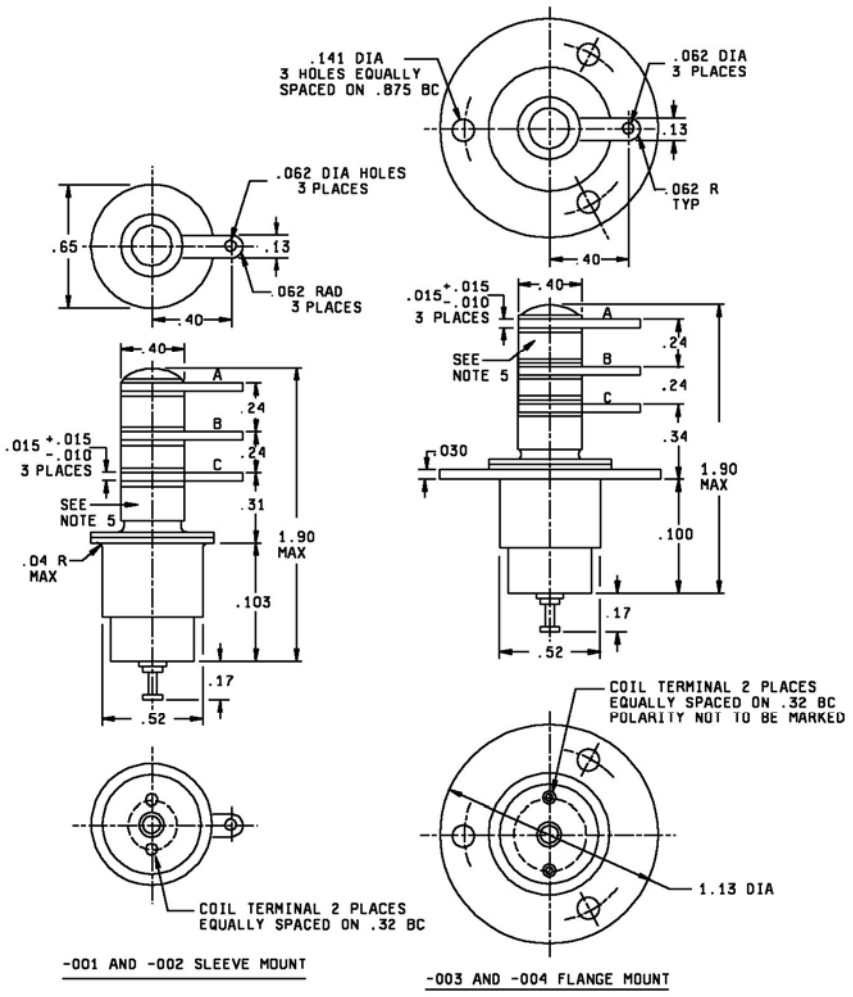
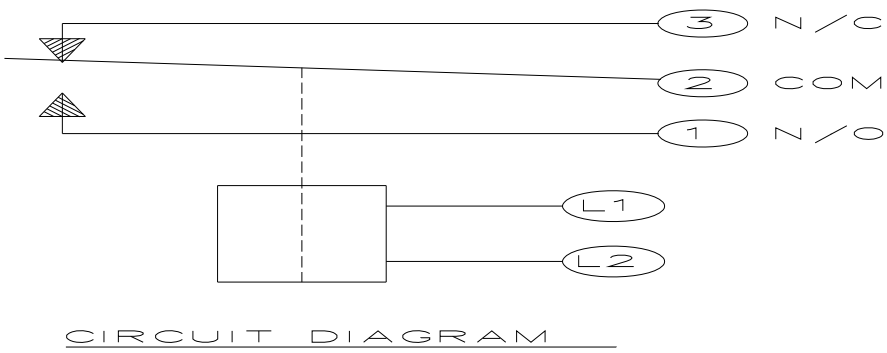



FIGURE 1. Relay, dimensions and configurations.

AMSC N/A

FSC 5945

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CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. Military part number shall be permanently marked on the relay coil housing.
5. N/O, N/C and Com to be stamped on plate below corresponding terminal.

FIGURE 1. Dimensions and configuration - Continued.

REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT, ground isolated.
Arrangement: 1 form C.
Load ratings:
Continuous carry, resistive:
75 amperes rms, dc or 60 Hz rms.
22 amperes rms, 2.5 MHz.
15 amperes rms, 16 MHz.

Make and break resistive:

9 kilovolts or 5 amps dc or 60 Hz maximum.
45 kilowatts maximum.

Voltage ratings, (continuous):
15 kilovolts peak, dc or 60 Hz.
13 kilovolts peak, 2.5 MHz.
10 kilovolts peak, 16 MHz.

Contact resistance:
Rated Life:
Before: .012 ohm, maximum.
During: .10 ohm, maximum.
After: .10 ohm, maximum.

Capacitance:
1.8 picofarads, maximum across open contacts.
1.5 picofarads, maximum between open contacts and ground.

COIL DATA:

Duty rating: Continuous.
Coil voltage: See table I.
Pickup voltage: See table I.
Dropout voltage: See table I.
Coil resistance: See table I.
Operate time: 30 milliseconds, maximum.
Release time: 10 milliseconds, maximum.



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ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage at atmospheric pressure:

Between all mated contacts in the open position, 20 kilovolts peak, 60 Hz or dc.

Between high voltage terminals and housing 20 kilovolts, 60 peak Hz or dc.

Between coil and housing: 500 V rms, 60 Hz.

ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.
Shock: MIL-STD-202-213, test condition J.

PHYSICAL:

Terminal strength: 5 pounds pull.
Dimensions and configuration: See figure 1.
Terminations: Screw
Weight: 16 ounces, maximum.

LIFE TEST REQUIREMENTS:

Load life: 20,000 cycles, 2 sample units, maximum cycling rate 20 per minute 9 kilowatts dc and 5 amperes dc make and break resistive. After test common to base, resistance shall be 500 megohms minimum.

Mechanical cycling: 100,000 cycles, 2 sample units (maximum cycling rate, 36,000 per hour; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage:
Tests to be conducted at atmospheric pressure rating only.
Duration of application: 5-10 seconds at a 10 percent increase in the dielectric withstanding voltage.

Part number: See table I.

TABLE I. Part number and characteristics.

| Military part number | Coil voltage V dc | Pick up voltage V dc max | Dropout voltage V dc | Coil resistance ±10% Ohms |
|---|---|---|---|---|
| M83725/13-001 | 6 | 4 | .2 -2.7 | 12 |
| M83725/13-002 | 12 | 9 | .5 - 5.8 | 52 |
| M83725/13-003 | 26.5 | 16 | 1 - 10 | 225 |
| M83725/13-004 | 48 | 33 | 1 - 20 | 735 |
| M83725/13-005 | 115 | 70 | 2 - 40 | 3900 |



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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-83725, this document references the following:

MIL-STD-202-213

Custodians:
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activity:
Air Force - 99

(Project 5945-2016-039)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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